OPTIMIZATION STUDY

CDR Baseline optimization candidate

Revision Final iterate from CDR_Baseline_optimization_test (2026-11-09) · plate-frame frame · x44-standard motors

Named input case · design_optimized.json

2026-11-09T23:00:43+00:00

CALCULATED route timing: BWR/motor-limited straight profiles and retained empirical point-turn equations; not hardware calibrated. Legacy traction/stability checks remain active. Deposition and relocation retain their reference timing.

Calculated route timing: traversal_time_one_way is inactive. Speed and acceleration inputs are upper bounds. Deposition and fresh-site relocation retain reference timing.

BWR soil source-range transfer flags: 3 across the two route legs. The published soil proxy is uncalibrated for this rover; see model provenance and the BWR documentation.

Actuator offsets are measured from the drum axle: positive side is outward along the arm; positive top is above the arm. The straight tube extension is included in mass and loads. The normal bracket is assumed rigid and massless; its mass, strength and interference remain unverified.

Actuator body packaging remains unverified. New PA-04 runs enforce the selected 6/8/9/10-inch command endpoints; the body/bracket collision reserve remains diagnostic.

NASA IPEx horizontal-force model. Outside published source conditions; the force is a provisional transfer. Tip/feed ratio: 10.935; source condition: 8.5. Source depth labels conflict with its tables. Shaft torque still uses the existing unvalidated force-times-radius assumption. Equal/opposite drum reactions are assumed to cancel; wheel feed work covers rolling and acceleration.

Constraints excluded from optimizer: empty_slope_traversal_margin, loaded_slope_traversal_margin, startup_position_margin.

These checks are still calculated and included in full model feasibility. Optimizer success with a failed excluded check is not a feasible design.

Optimizer did not converge successfully

73 / 76 physical model checks pass · 7 capabilities not assessed

Predictions are conditional on the failed checks below. These are predictions from the stated inputs, not measured performance.

Dry rover mass

41.5 kg

Before adding regolith

Payload per cycle

116.4 kg

Total carried by four drums

Complete cycles

3

Within 15 minutes; no partial delivery credit

Delivered volume

209.5 L

Complete-cycle mission estimate

Construction score

Continuous mission estimate using the configured mass and energy weights. The complete-cycle estimate is reported in Performance & mass.

Construction points

2,142.49

Weighted mass + energy points

Mass points

1,532.77

Dry-mass productivity component

Energy points

609.72

Consumed-energy productivity component

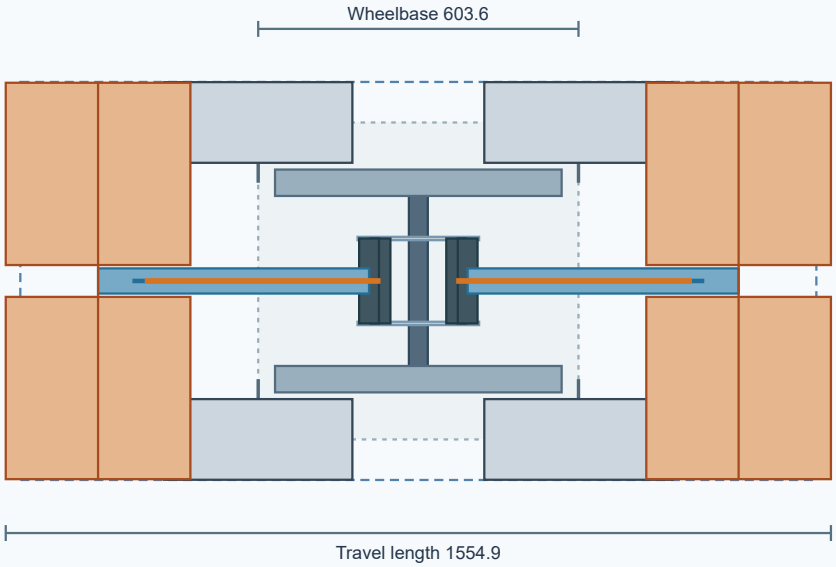
01 / ROVER CONFIGURATION

Geometry at the evaluated inputs

Level-ground schematics and one prescribed obstacle pose. Check statuses below determine whether envelopes and clearance requirements are satisfied.

Top-down view / travel pose

Orthographic footprints; front is +x (right). Dimensions in mm.



Wheel-center track: 597.2 | Overall width: 749.2 | Wheel width: 152.0

Each drum: 347.9 diameter x 343.5 axial length | Pair gap: 60.0

Blue: one arm per drum pair · orange: actuators · charcoal: aluminum cross-tubes · slate: rockers and axle.

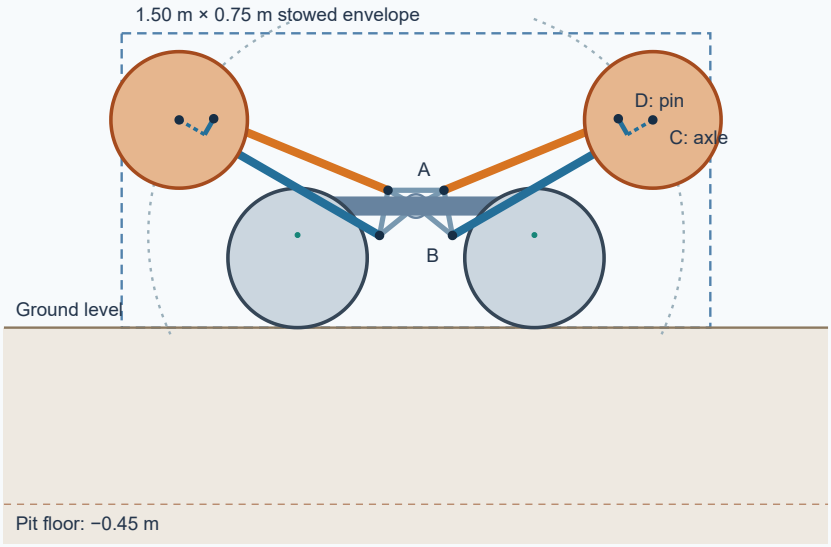
Four aluminum cross-tubes: upper/lower footprints can overlap. Dashed blue: 1500 x 750 stowed limit.

Plate outlines are schematic. Projection overlap does not prove collision. Heights and equipment are not shown.

Travel-pose plan view. Four wheels and four drums are shown at their model stations. Top-down SVG

Travel

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



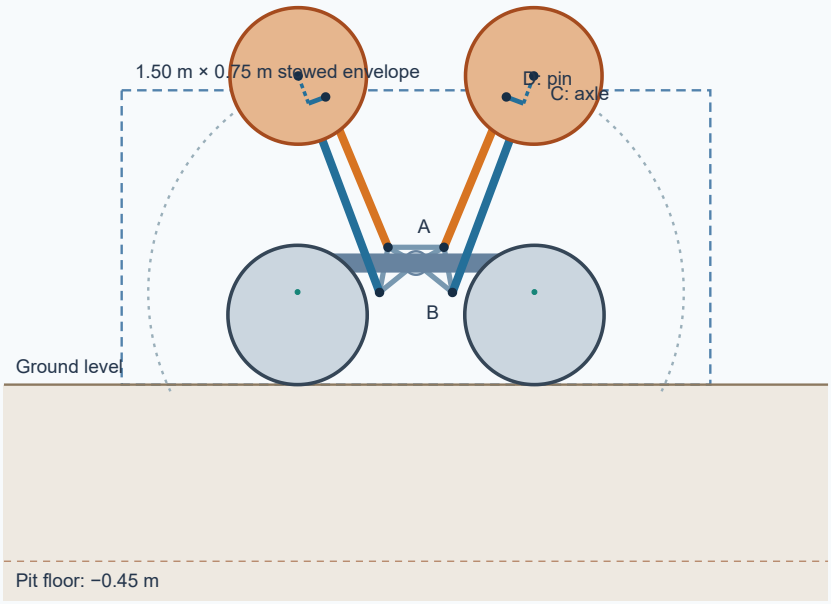
Actuator: 480.2 mm · Fixed arm: 589.4 mm · Drum Ø 347.9 mm

Drum top: +0.703 m · Drum bottom: +0.355 m · A: 0.350 m · B: 0.235 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -74.7 mm / top +46.6 mm · Ground is y = 0

Deposition · both pairs

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



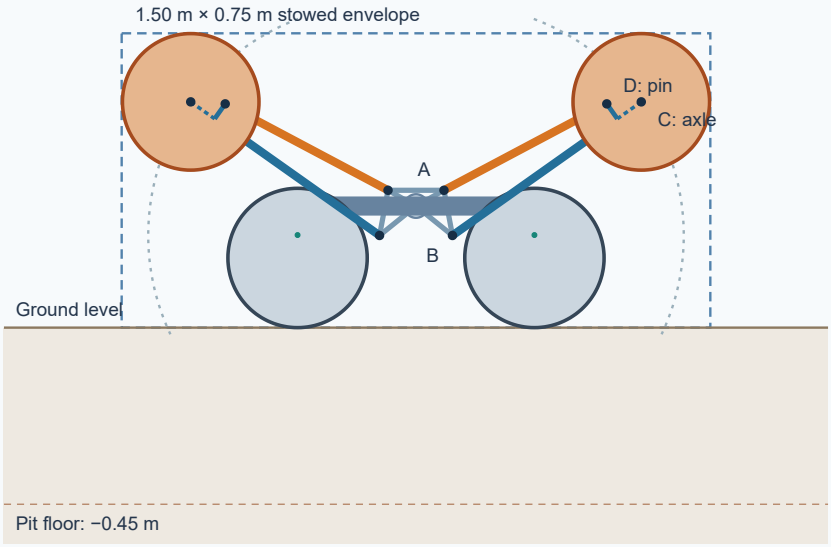
Actuator: 414.8 mm · Fixed arm: 589.4 mm · Drum Ø 347.9 mm

Drum top: +0.961 m · Drum bottom: +0.613 m · A: 0.350 m · B: 0.235 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -74.7 mm / top +46.6 mm · Ground is y = 0

Compact · starting envelope

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



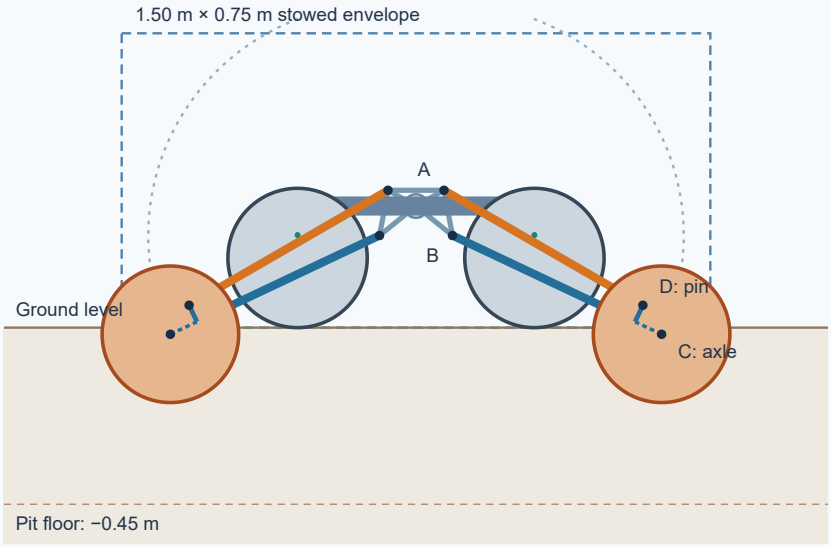
Actuator: 469.7 mm · Fixed arm: 589.4 mm · Drum Ø 347.9 mm

Drum top: +0.749 m · Drum bottom: +0.401 m · A: 0.350 m · B: 0.235 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -74.7 mm / top +46.6 mm · Ground is y = 0

Maximum extension · plunge / recovery reach

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



Actuator: 585.5 mm · Fixed arm: 589.4 mm · Drum Ø 347.9 mm

Drum top: +0.157 m · Drum bottom: -0.191 m · A: 0.350 m · B: 0.235 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -74.7 mm / top +46.6 mm · Ground is y = 0

Blue: arm · orange: actuator. Each visible drum represents a shaft-linked pair. Drawings are not CAD.

Travel and deposition mass properties

Pose / loading	Longitudinal CG	CG height	Yaw inertia
Travel · fully loaded	0.00 mm	446.5 mm	57.9 kg·m²
Deposition · fully loaded	0.00 mm	662.3 mm	21.8 kg·m²
Deposition turn · one pair full	175.2 mm	623.6 mm	13.2 kg·m²
Deposition · empty	-1.07e-14 mm	476.2 mm	4.55 kg·m²

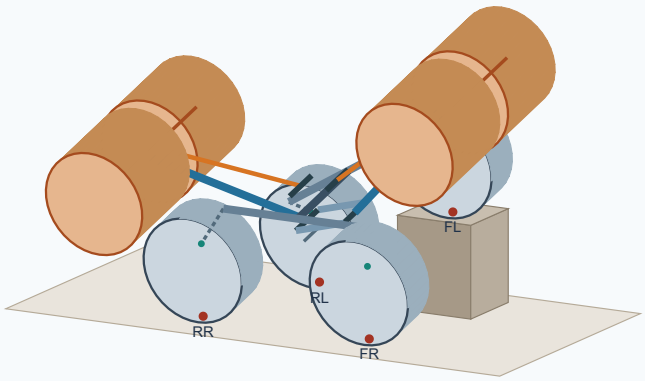
Added pose-change duration: 13.2 s. Added electrical energy: 4,349 J. Both pairs move together while full before dumping, then return empty after both dumps.

Raising reduces yaw inertia only when the drums move inward. Higher CG can reduce dynamic stability. One wheelbase must satisfy both travel and deposition checks. Half-loaded arm repositioning and dynamic dumping require controller and hardware validation.

One wheel on the rock / travel pose

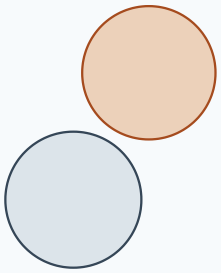
Front-left raised: 300.0 mm · chassis pitch 15.3° · roll 12.8°

Same kinematic pose as the model; arms held in travel position.



Common roll removed

Closest cylinder pair: drum 1 / FL



Radial circle gap: +33.88 mm

Axial end gap: -150.86 mm

Finite-cylinder gap: +33.88 mm

Required clearance: 15.00 mm

One centered arm/actuator per drum pair · blue: arms · orange: actuators · charcoal: aluminum cross-tubes · gray: wheels · tan: drums.

Support block shows the prescribed height; its shape and width are illustrative. Actual rock climbing is not simulated.

Contact model omits camber effects. The right section also needs the axial-width check; circles alone can over-reject.

Parallel-cylinder clearance is the default model. This section shows the closest pair for the illustrated front-left raise.

Plate-frame member schematic

Plate frame: provisional member topology

Lengths and directions follow the mounting coordinates; this is not the manufactured cutout outline.



Plate thickness: 6.35 mm (fixed)

Common strip width: 13.32 mm

cross-tube ID / OD: 31.75 / 37.30 mm

Plate spacing / cross-tube span: 160.0 mm

Rocker bearing spacing: 370.1 mm

Wheel-to-rocker gap: 12.7 mm

Aluminum arm box: 47.43 x 48.27 mm

Arm wall: 1.0000 mm

Central support: pin, balanced mirrored gravity only.

cross-tube material properties are PROVISIONAL.

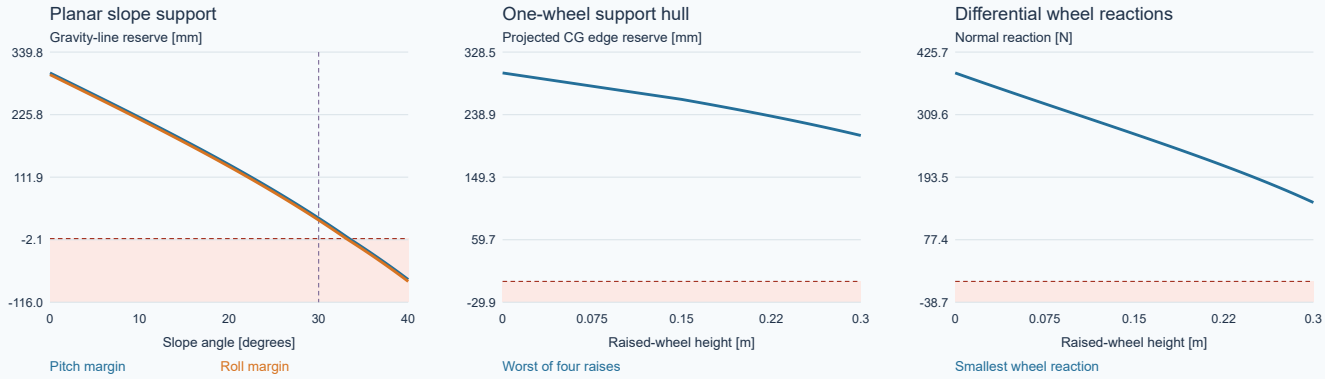
Spoke buckling is a strip screen,

not whole-plate buckling validation.

Static slope and articulation curves

Loaded travel pose: gravity stability screens

Linear springs + hard stops. Curves after wheel unloading are guarded continuations; slope and rock are separate cases.



LINKAGE / ACTUATOR LOADS

Actuator force versus stroke



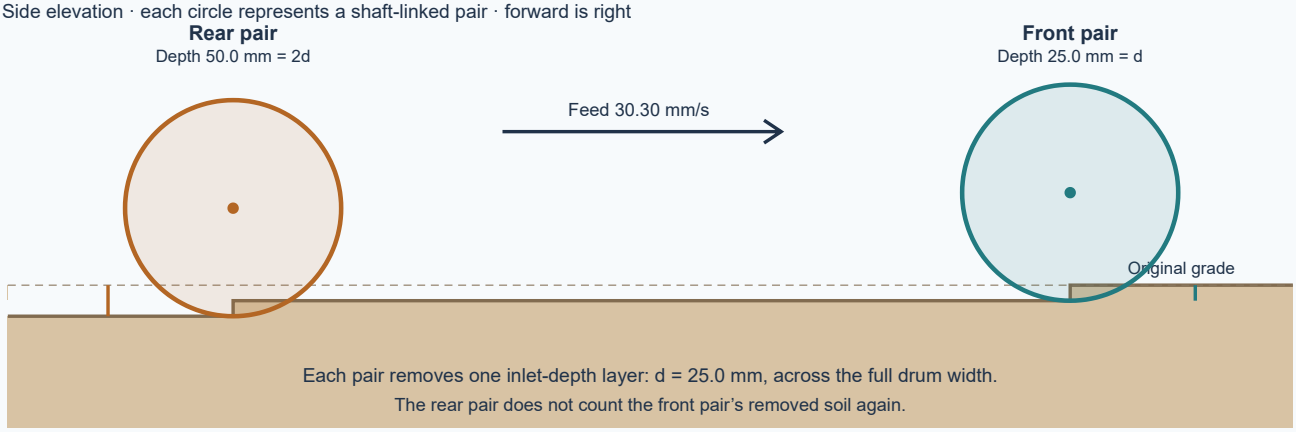
EXCAVATION / FORWARD FEED

Drive straight with the drums lowered

Each pair cuts one inlet-depth fresh layer across its full drum width. Radial protrusion equals inlet depth; the rear cuts a second layer in the same tracks. Forward advance per scoop remains limited by inlet depth.

Moving excavation · two fresh layers

Side elevation · each circle represents a shaft-linked pair · forward is right



Both pairs lower together → feed with both pairs cutting → both pairs raise together.

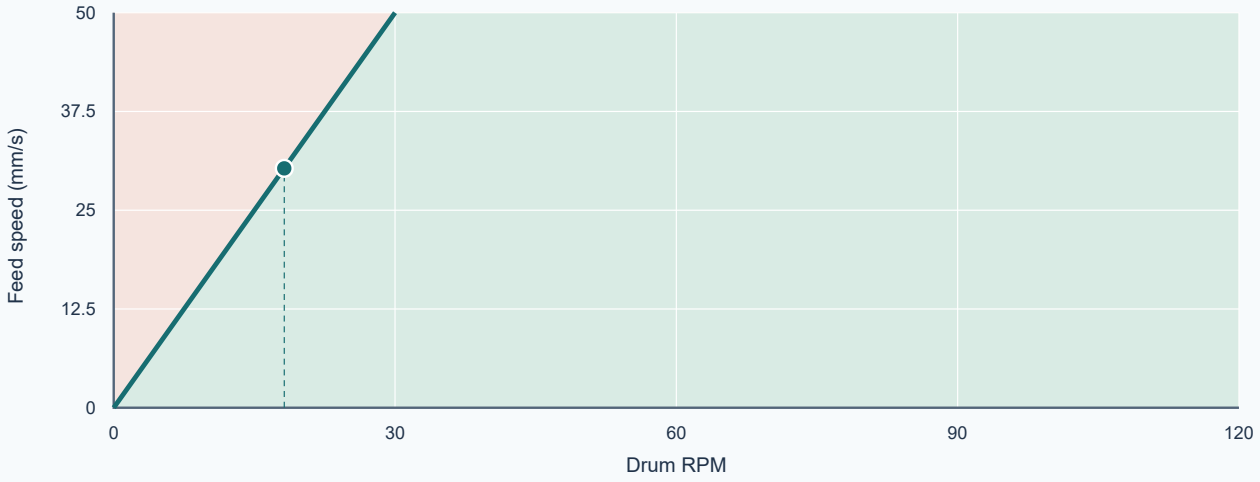
Productive lane: 3.475 m · total feed travel: 4.823 m · feed time: 159.4 s

The maximum-extension view is a separate plunge/recovery reach screen; it is not this operating pose.

Schematic soil boundary; entry/endcap soil is not credited. Soil flow and recovery driving require validation.

Feed speed and drum RPM

Forward advance per scoop ≤ inlet depth · an uncalibrated feed screen



Current: 18.19 rpm, 30.30 mm/s

Forward advance: 24.99 mm/scoop · inlet depth: 25.00 mm

Maximum feed = inlet_depth × scoops/revolution × RPM / 60; at this RPM: 30.31 mm/s.

Green: passes this feed screen. Red: excessive advance per scoop. Motor, reach and traction checks still apply.

This volume/bite constraint does not establish measured capture, cutting loads or high-speed soil retention.

Excavation quantity	Evaluated value
Inlet depth / fresh-layer cut depth	25.0 mm
Inlet width (full drum width)	343.5 mm
Front depth below original grade	25.0 mm
Rear depth below original grade	50.0 mm
Straight excavation feed speed	30.3 mm/s
Excavation drum RPM	18.2 rpm
Advance per scoop	25.0 mm
Feed limit at selected drum RPM	30.3 mm/s
Productive excavation lane length	3,475 mm
Feed travel including tandem staging	4,823 mm
Straight-feed duration	159.4 s
Feed-drive electrical energy	1,537 J
Feed-drive peak electrical power	13.3 W

Excavation quantity	Evaluated value
Plunge/recovery: axle reaches ground level	17.0 mm
Lower both arms together	8.17 s
Raise both arms together	8.09 s
Total excavation arm motion	16.3 s
Excavation arm electrical energy · both actuators	5,229 J

Feed-drive electrical work and peak power are included in digging totals. The old stationary-plunge soil-volume calculation is inactive. Maximum extension remains a separate capability envelope. Reaching the ground with an axle does not verify motor clearance or driving on the drums.

02 / PREDICTED OUTPUT

Performance & mass

Complete-cycle estimates and continuous MDO estimates are labeled separately.

Prediction	Value	Meaning
Full cycle duration	290.0 s	Includes digging, actuation, travel, dumping and relocation; overlap removed once.
Excavation including arm motion	183.7 s	Both arms lower together, feed/cut, then both raise together.
Deposition including arm motion and maneuver	27.6 s	Position both arms, dump, turn, dump, and return arms to travel.
Excavation + deposition	211.3 s	Excludes route, fresh-site relocation and alignment overhead.
Energy per cycle	11.5 Wh	Electrical energy including auxiliaries.
Mission energy · complete cycles	34.6 Wh	Includes idle electronics for remaining mission time.
Mission energy · continuous estimate	35.5 Wh	Used by the existing battery constraint and MDO objective.
Peak electrical power	635.7 W	Largest modeled simultaneous demand; not an average power rating.
Loaded mass	157.8 kg	Dry mass + all four drums' regolith payload.
Traversal timing mode	calculated	BWR/motor-limited candidate profiles; legacy turn screens remain enforced.
Loaded outbound route	26.0 s	Includes route turns; excludes deposition and relocation.
Empty return route	26.0 s	Includes route turns; excludes deposition and relocation.
Actual route peak speed	1.00 m/s	Calculated profile, not measured vehicle performance.
Score · complete-cycle estimate	2,087.45 points	Uses complete deliveries and idle remainder; separate from the continuous MDO objective.

Prediction	Value	Meaning
Empty rover · 30° climb	No climbing solution	BWR, GRC-1, equal mean wheel loads. Force shortfall per wheel: 30.34 N. Acceptance: solved slip strictly below 1. Soil-fit range flags: 0.
Full rover · 20° climb	No climbing solution	BWR, GRC-1, equal mean wheel loads. Force shortfall per wheel: 45.60 N. Acceptance: solved slip strictly below 1. Soil-fit range flags: 3.

Dry mass breakdown

Each assembly is counted once.

Equipment allowance	**6.00 kg**
Frame	**4.66 kg**
Four bare wheels	**5.88 kg**
Four drive assemblies	**4.15 kg**
Four drums + two drive assemblies	**13.5 kg**
Battery	**1.58 kg**
Arms + actuators	**5.71 kg**

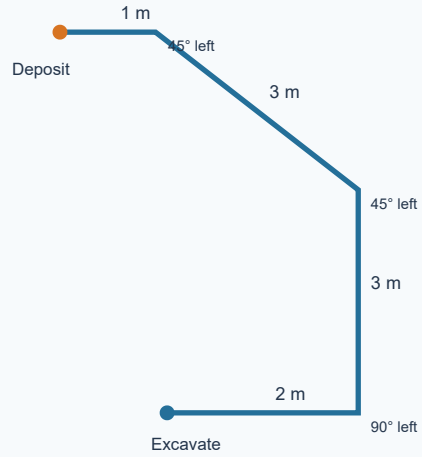
Electrical energy per cycle

Work / provisional efficiency, plus auxiliaries.

Digging	**4.64 Wh**
Dumping	**0.15 Wh**
Travel + deposition turn	**1.92 Wh**
Excavation actuators	**1.45 Wh**
Deposition pose changes	**1.21 Wh**
Fresh-site move	**0.07 Wh**
Electronics	**2.01 Wh**

Route & cycle accounting

26.029 s outbound + 26.029 s return



Return follows the same path in reverse order; yaw signs reverse.

26.029 s outbound + 26.029 s return

Lower both pairs together → drive the excavation pass → raise both pairs together → outbound → move both pairs to deposition → dump one pair → turn → dump second pair → return both pairs to travel → return route → relocate to a fresh lane.

Each excavation arm phase lasts as long as the slower arm needs at its actual digging position. Lowering and cutting do not overlap. Maximum extension remains a separate load/capability check. The deposition turn and fresh-site relocation are additional to the timed route. Modeled full cycle: 290.0 s.

03 / GAPS TO RESOLVE

Requirements & model checks

Failures are listed first. “Not assessed” means the model does not verify that capability.

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Empty rover: 30° BWR climb (excluded from optimizer) `empty_slope_traversal_margin`	No climbing solution `Slip = 1 is a failure sentinel`	Slip < 1 (strict)	-0.34	Fail
Full rover: 20° BWR climb (excluded from optimizer) `loaded_slope_traversal_margin`	No climbing solution `Slip = 1 is a failure sentinel`	Slip < 1 (strict)	-0.13	Fail
Startup position margin (excluded from optimizer) `startup_position_margin`	Reserve: -10.5 mm	Reserve ≥ 0	-10.5 mm	Fail
Dry rover mass `dry_mass`	41.5 kg	≤ 80.0 kg	38.5 kg	Pass
Compact overall length `rover_length`	1,496 mm	≤ 1,500 mm	3.95 mm	Pass
Overall width `rover_width`	749.2 mm	≤ 750.0 mm	0.75 mm	Pass
Compact overall height `rover_height`	749.3 mm	≤ 750.0 mm	0.75 mm	Pass
Ground clearance: lower axle after wheel/bar deductions `ground_clearance_margin`	Reserve: 2.88 mm	Reserve ≥ 0	2.88 mm	Pass
Excavation shaft-power reserve `drum_power_margin`	Reserve: 1,189 W	Reserve ≥ 0	1,189 W	Pass
Excavation: penetration reserve `penetration_margin`	Reserve: 314.5 N	Reserve ≥ 0	314.5 N	Pass
Inlet depth: radial geometry reserve `cut_depth_margin`	Reserve: 18.5 mm	Reserve ≥ 0	18.5 mm	Pass
Ground contact pressure `ground_pressure_margin`	17.9 kPa	≤ 25.0 kPa	7.12 kPa	Pass
Straight travel: motor-force reserve `straight_motor_margin`	Reserve: 412.6 N	Reserve ≥ 0	412.6 N	Pass
Turning: motor-force reserve `turn_motor_margin`	Reserve: 5,323 N	Reserve ≥ 0	5,323 N	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Straight travel: traction reserve `straight_traction_margin`	Reserve: 169.6 N	Reserve ≥ 0	169.6 N	Pass
Turning: traction reserve `turn_traction_margin`	Reserve: 0.30 N	Reserve ≥ 0	0.30 N	Pass
Straight travel: shaft-power reserve `straight_power_margin`	Reserve: 2,239 W	Reserve ≥ 0	2,239 W	Pass
Turning: shaft-power reserve `turn_power_margin`	Reserve: 2,515 W	Reserve ≥ 0	2,515 W	Pass
Battery usable energy `battery_energy_margin`	35.5 Wh	≤ 176.5 Wh	141.0 Wh	Pass
Battery peak-power capacity `battery_power_margin`	635.7 W	≤ 1,471 W	835.5 W	Pass
Time remaining after one complete cycle `complete_cycle_margin`	Reserve: 610.0 s	Reserve ≥ 0	610.0 s	Pass
Pit depth margin `pit_depth_margin`	Reserve: 259.1 mm	Reserve ≥ 0	259.1 mm	Pass
Transport clearance margin `transport_clearance_margin`	Reserve: 321.3 mm	Reserve ≥ 0	321.3 mm	Pass
Travel drum height above wheel-diameter step `wheel_first_margin`	Reserve: 0.36 mm	Reserve ≥ 0	0.36 mm	Pass
Deposition pose: triangle reach reserve `deposition_reach_margin`	Reserve: 0.10 mm	Reserve ≥ 0	0.10 mm	Pass
Deposition command below maximum extension `deposition_position_margin`	Reserve: 170.7 mm	Reserve ≥ 0	170.7 mm	Pass
Deposition drum above ground `deposition_clearance_margin`	Reserve: 532.7 mm	Reserve ≥ 0	532.7 mm	Pass
Dig position margin `dig_position_margin`	Reserve: 115.8 mm	Reserve ≥ 0	115.8 mm	Pass
Pivot spacing margin `pivot_spacing_margin`	Reserve: 37.2 mm	Reserve ≥ 0	37.2 mm	Pass
Pivot mount margin `pivot_mount_margin`	Reserve: 355.0 mm	Reserve ≥ 0	355.0 mm	Pass
Pivot longitudinal margin `pivot_longitudinal_margin`	Reserve: 208.9 mm	Reserve ≥ 0	208.9 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Actuator pivot longitudinal margin `actuator_pivot_longitudinal_margin`	Reserve: 230.5 mm	Reserve ≥ 0	230.5 mm	Pass
Bottom mount margin `bottom_mount_margin`	Reserve: 57.3 mm	Reserve ≥ 0	57.3 mm	Pass
Stroke margin `stroke_margin`	Reserve: 65.3 mm	Reserve ≥ 0	65.3 mm	Pass
Triangle retracted margin `triangle_retracted_margin`	Reserve: 65.5 mm	Reserve ≥ 0	65.5 mm	Pass
Triangle extended margin `triangle_extended_margin`	Reserve: 33.6 mm	Reserve ≥ 0	33.6 mm	Pass
Toggle margin `toggle_margin`	Reserve: 192.3 mm	Reserve ≥ 0	192.3 mm	Pass
Actuator toggle margin `actuator_toggle_margin`	Reserve: 211.9 mm	Reserve ≥ 0	211.9 mm	Pass
Wheel clearance margin `wheel_clearance_margin`	Reserve: 0.30 mm	Reserve ≥ 0	0.30 mm	Pass
Deployed height margin `deployed_height_margin`	Reserve: 1,539 mm	Reserve ≥ 0	1,539 mm	Pass
Actuator force capacity `actuator_force_margin`	3,826 N	≤ 4,003 N	177.7 N	Pass
Excavation torque reserve `drum_torque_margin`	Reserve: 260.5 N·m	Reserve ≥ 0	260.5 N·m	Pass
Dumping torque reserve `dump_torque_margin`	Reserve: 87.5 N·m	Reserve ≥ 0	87.5 N·m	Pass
Minimum payload reserve `payload_margin`	Reserve: 116.3 kg	Reserve ≥ 0	116.3 kg	Pass
Excavated trench diameter reserve `trench_traverse_margin`	Reserve: 305.0 mm	Reserve ≥ 0	305.0 mm	Pass
Fresh site margin `fresh_site_margin`	Reserve: 53.0 mm	Reserve ≥ 0	53.0 mm	Pass
Frame section margin `frame_section_margin`	Reserve: 7.80 mm	Reserve ≥ 0	7.80 mm	Pass
Arm section margin `arm_section_margin`	Reserve: 10.9 mm	Reserve ≥ 0	10.9 mm	Pass
Structural stress envelope `structure_stress_margin`	119.9 MPa	≤ 120.0 MPa	0.10 MPa	Pass
Structural deflection envelope `structure_deflection_margin`	2.65 mm	≤ 5.00 mm	2.35 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Structural buckling reserve `structure_buckling_margin`	Reserve: 2,938 N	Reserve ≥ 0	2,938 N	Pass
Level-route dynamic tip reserve `mobility_stability_margin`	Reserve: 281.6 mm	Reserve ≥ 0	281.6 mm	Pass
Asymmetric dump tip reserve `tip_stability_margin`	Reserve: 101.5 mm	Reserve ≥ 0	101.5 mm	Pass
Rock pose: geometric reach reserve `obstacle_pose_margin`	Reserve: 280.6 mm	Reserve ≥ 0	280.6 mm	Pass
Rock endpoint: global tip reserve `obstacle_tip_margin`	Reserve: 209.0 mm	Reserve ≥ 0	209.0 mm	Pass
Obstacle drum–wheel clearance `obstacle_drum_wheel_margin`	33.9 mm	≥ 15.0 mm	18.9 mm	Pass
30° slope: static tip reserve `slope_tip_margin`	Reserve: 33.4 mm	Reserve ≥ 0	33.4 mm	Pass
Rock path: global tip reserve `rock_sweep_tip_margin`	Reserve: 209.0 mm	Reserve ≥ 0	209.0 mm	Pass
Rock path: smallest wheel reaction `rock_sweep_wheel_load_margin`	Reserve: 146.2 N	Reserve ≥ 0	146.2 N	Pass
Available excavation area `excavation_area_margin`	Reserve: 0.01 m^2	Reserve ≥ 0	0.01 m^2	Pass
Fresh-site move: drive/traction reserve `site_drive_margin`	Reserve: 156.7 N	Reserve ≥ 0	156.7 N	Pass
Plate geometry reserve `plate_geometry_margin`	Reserve: 7.04 mm	Reserve ≥ 0	7.04 mm	Pass
Frame packaging reserve `frame_packing_margin`	Reserve: 91.9 mm	Reserve ≥ 0	91.9 mm	Pass
Plate strip stress `plate_stress_margin`	96.3 MPa	≤ 120.0 MPa	23.7 MPa	Pass
Plate strip buckling reserve `plate_buckling_margin`	Reserve: 2,938 N	Reserve ≥ 0	2,938 N	Pass
aluminum cross-tube bending stress `cross_tube_bending_margin`	94.9 MPa	≤ 120.0 MPa	25.1 MPa	Pass
Slope: minimum normal wheel reaction `slope_wheel_load_margin`	Reserve: 0.50 N	Reserve ≥ 0	0.50 N	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Static wheel equilibrium and regular-branch reserve `wheel_equilibrium_margin`	Reserve: 0.85	Reserve ≥ 0	0.85	Pass
Actuator catalog stroke domain `actuator_stroke_domain_margin`	Reserve: 152.4 mm	Reserve ≥ 0	152.4 mm	Pass
Actuator load-curve domain `actuator_curve_margin`	Reserve: 177.7 N	Reserve ≥ 0	177.7 N	Pass
Forward advance per scoop: inlet-depth reserve `feed_margin`	25.0 mm	≤ 25.0 mm	0.01 mm	Pass
Moving excavation: operating-pose reach `operating_reach_margin`	Reserve: 24.4 mm	Reserve ≥ 0	24.4 mm	Pass
Plunge/recovery: axle reaches ground level `axle_ground_margin`	-17.0 mm	≤ 0.00 mm	17.0 mm	Pass
Moving excavation: geometry domain `excavation_domain_margin`	Reserve: 149.0 mm	Reserve ≥ 0	149.0 mm	Pass
Excavation feed: motor-force reserve `feed_motor_margin`	Reserve: 7,135 N	Reserve ≥ 0	7,135 N	Pass
Excavation feed: shaft-power reserve `feed_power_margin`	Reserve: 2,669 W	Reserve ≥ 0	2,669 W	Pass
Lin act side arm offset search range `lin_act_side_arm_offset_bounds`	-74.7 mm	-500.0 mm to -50.8 mm	23.9 mm	Pass
Lin act top arm offset search range `lin_act_top_arm_offset_bounds`	46.6 mm	-150.0 mm to 150.0 mm	103.4 mm	Pass
Drum outer diameter search range `drum_diameter_bounds`	347.9 mm	160.0 mm to 359.0 mm	11.1 mm	Pass
Width of each drum search range `drum_length_bounds`	343.5 mm	120.0 mm to 359.0 mm	15.5 mm	Pass
Excavation drum RPM search range `drum_rpm_bounds`	18.2 rpm	5.00 rpm to 120.0 rpm	13.2 rpm	Pass
Straight excavation feed speed search range `feed_speed_bounds`	0.03 m/s	0.0001 m/s to 0.50 m/s	0.03 m/s	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Dumping speed search range `dump_rpm_bounds`	59.8 rpm	5.00 rpm to 60.0 rpm	0.22 rpm	Pass
Drum reduction ratio search range `drum_gear_ratio_bounds`	59.7	9.00 to 125.0	50.7	Pass
Inlet depth / fresh-layer cut depth search range `inlet_depth_bounds`	25.0 mm	3.00 mm to 25.0 mm	0.00 mm	Pass
Arm wall thickness search range `arm_wall_bounds`	1.00 mm	1.00 mm to 5.00 mm	0.00 mm	Pass
Battery capacity search range `battery_capacity_bounds`	220.7 Wh	10.0 Wh to 300.0 Wh	79.3 Wh	Pass
Travel actuator length search range `travel_actuator_length_bounds`	480.2 mm	403.1 mm to 606.3 mm	77.1 mm	Pass
Deposition actuator length search range `deposition_actuator_length_bounds`	414.8 mm	403.1 mm to 606.3 mm	11.7 mm	Pass
Compact actuator length search range `actuator_retracted_length_bounds`	469.7 mm	403.1 mm to 606.3 mm	66.6 mm	Pass
Maximum-extension actuator length search range `actuator_extended_length_bounds`	585.5 mm	403.1 mm to 606.3 mm	20.8 mm	Pass
Actuator pivot height search range `top_pivot_height_bounds`	350.0 mm	350.0 mm to 700.0 mm	0.00 mm	Pass
Actuator pivot longitudinal position search range `actuator_pivot_x_bounds`	71.3 mm	20.0 mm to 300.0 mm	51.3 mm	Pass
Arm pivot height search range `bottom_pivot_height_bounds`	234.8 mm	90.0 mm to 300.0 mm	65.2 mm	Pass
Arm length search range `arm_length_bounds`	589.4 mm	300.0 mm to 700.0 mm	110.6 mm	Pass
Arm pivot longitudinal position search range `pivot_x_bounds`	92.9 mm	50.0 mm to 400.0 mm	42.9 mm	Pass
Wheelbase search range `wheelbase_bounds`	603.6 mm	380.0 mm to 650.0 mm	46.4 mm	Pass
Wheel-center track search range `track_width_bounds`	597.2 mm	500.0 mm to 699.0 mm	97.2 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Aluminum arm box width search range `arm_box_width_bounds`	47.4 mm	25.0 mm to 100.0 mm	22.4 mm	Pass
Aluminum arm box height search range `arm_box_height_bounds`	48.3 mm	25.0 mm to 100.0 mm	23.3 mm	Pass
aluminum cross-tube span / plate spacing search range `cross_tube_span_bounds`	160.0 mm	160.0 mm to 460.0 mm	0.00 mm	Pass
Aluminum cross-tube wall thickness search range `cross_tube_wall_bounds`	2.77 mm	1.00 mm to 6.00 mm	1.77 mm	Pass
Plate strip width search range `plate_spoke_width_bounds`	13.3 mm	12.0 mm to 70.0 mm	1.32 mm	Pass
Rocker box width search range `rocker_box_width_bounds`	49.7 mm	20.0 mm to 65.0 mm	15.3 mm	Pass
Rocker box height search range `rocker_box_height_bounds`	48.8 mm	25.0 mm to 90.0 mm	23.8 mm	Pass
Rocker wall search range `rocker_wall_bounds`	1.01 mm	1.00 mm to 5.00 mm	0.01 mm	Pass
Axle outer diameter search range `axle_outer_diameter_bounds`	35.6 mm	25.0 mm to 90.0 mm	10.6 mm	Pass
Axle wall search range `axle_wall_bounds`	1.11 mm	1.00 mm to 6.00 mm	0.11 mm	Pass
Rocker pivot above wheel centers search range `rocker_pivot_offset_bounds`	132.3 mm	20.0 mm to 150.0 mm	17.7 mm	Pass
Wheel diameter search range `wheel_diameter_bounds`	355.0 mm	180.0 mm to 359.0 mm	4.00 mm	Pass
Wheel width search range `wheel_width_bounds`	152.0 mm	50.0 mm to 359.0 mm	102.0 mm	Pass
Drum-supported recovery driving	Axle-to-ground reach is screened geometrically. Drum traction, motor-ground clearance, recovery loads and actual escape capability require hardware and soil tests.	—	—	Not assessed

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Slope motor limits / sideways sliding	The separate BWR climb checks accept slip<1 only. Motor capability on slopes, lateral sliding and controller behavior are not evaluated.	—	—	Not assessed
Real 30 cm rock traversal	Static poses and clearance screens do not establish the driving forces or actual terrain-contact sequence.	—	—	Not assessed
50 cm ditch crossing	Ditch width/radius and contact sequence have not been defined.	—	—	Not assessed
Continuous motor operation	Stator-current limits and thermal duty remain undecided.	—	—	Not assessed
Connections & material validation	Bonded inserts, bushings, bolts, local failure and selected aluminum cross-tube properties need separate checks.	—	—	Not assessed
Soil and skid-steer predictions	Rolling, traction, capture and scrub relationships remain uncalibrated.	—	—	Not assessed

Positive margins satisfy the modeled criterion. Status uses the saved model's numerical tolerance. Search ranges are hidden by default and available in the filter; they remain part of saved feasibility. Margins use their own units and cannot be ranked directly across unlike checks.

04 / ENGINEERING LOADS

Loads & structural response

Values identify their load case and whether they apply per component or to the whole rover.

Frame gravity solution is valid for its declared load case. Structural load factor: 1.5; structural safety factor: 2.

Quantity / scope	Prediction	Load case / interpretation
Required force per actuator `required_actuator_force`	3,826 N	Quasi-static gravity envelope for loaded drums and arm; extraction resistance and dynamic loads are not modeled.
Frame gravity reaction at each actuator `frame_gravity_actuator_force`	5,739 N	Mirrored loaded gravity sweep; includes the structural load factor. Separate from actuator sizing load.
Nominal force per wheel `frame_nominal_wheel_reaction`	387.0 N	Equal-share level-ground weight. Not a rock or slope wheel-load envelope.
Total cutting force `cut_force`	184.4 N	Across four drums at the selected bite and engagement.
Worst modeled contact pressure `ground_pressure`	17.9 kPa	Route pressure estimate, including load transfer and rocker factor.
Structural stress envelope `structure_stress`	119.9 MPa	Envelope of local member stresses; not one assembled-frame FEA result.
Structural deflection envelope `structure_deflection`	2.65 mm	Conservative combination of local/relative deflections.
Upper/lower mount relative displacement `frame_mount_relative_deflection`	0.91 mm	Frame gravity sweep.
Rocker-end flexure `rocker_deflection`	0.97 mm	Local beam-screen displacement.
Suspension-axle flexure `axle_deflection`	0.34 mm	Axle-overhang screen.
Arm-tip flexure `arm_deflection`	2.65 mm	Relative arm beam displacement.
Loaded CG height `loaded_cg_z`	446.5 mm	Fully loaded travel pose, above level ground.

Model notes & engineering data

Keep assumptions and validity limits attached to the predictions.

Obstacle and structural interpretation

Requested endpoint rock pose is geometrically reachable; this alone does not establish traversal. Static wheel reactions and body/rocker pose are solved together with linear springs and a 64.0 mm rigid compression stop. Stop contact can carry extra reaction; forces are not capped. Negative normal reactions reject the four-contact pose. The global support limit and first wheel unloading are separate diagnostics. After unloading or an invalid solve, continued geometry is guarded and is not a predicted post-lift trajectory. The heading search uses a 5-degree grid with bounded refinement; it is a numerical search, not a proof over all headings. Finite contact patches, wheel camber corrections, lateral compliance and soil sinkage are unmodeled. Wheel stiffness and the stop are student assumptions pending FEA and physical tests. Geometry/clearance schematics remain nominal.

Obstacle drum–wheel clearance

Requested-height obstacle clearance screen. Parallel finite cylinders: shared roll is removed, then circular-plane and axial-width gaps are combined. Negative gap means overlap of the ideal solids. A common roll rotation preserves relative clearance in both models; rocker articulation still changes relative geometry.

Ditch requirement remains unverified. Static slope tipping assumes adequate friction; it does not verify climbing. The 5 mm structural limit applies to the modeled envelope of local deflections, not one assembled-frame displacement.

For the plate frame, aluminum cross-tube stiffness/strength are PROVISIONAL. Connections, dynamic obstacle loads, cutting/recovery frame cases and whole-plate buckling need separate verification.

Standard X44 motors · assumptions

Four wheel motors at 100:1; two drum motors at 59.67:1 (exploratory gearing).

Electrical energy = output-shaft mechanical work / combined efficiency. Drive: 72%; drums: 70%. Both efficiencies are provisional.

Continuous operation unverified. Current-limit inputs of zero mean undecided. Shaft torque-speed capability is enforced separately from energy accounting.

PA-04 actuators · 900 lbf / 12 V

Two actuators. Continuous catalog stroke: 203.2 mm; installed hardware limits: 403.1 mm to 606.3 mm. These include 0.00 mm added to each endpoint. Compact, travel, deposition and digging commands are compared with these limits diagnostically; packaging is not an active constraint.

Mass per actuator: 2.56 kg. Raising envelope: 0.01 m/s and 13.6 A; maximum-reach diagnostic times: raising 10.5 s, lowering 10.6 s. Lowering assumes rated-load speed and 14 A. Electrical energy uses 12 V × current × time for both actuators, including motor losses.

Mass fit supports 2–40 inch strokes; load curves support 0–900 lbf. Guarded out-of-domain diagnostics retain failed checks. Independent hardware, gravity-assisted lowering, duty cycle, temperature and installation-hardware mass remain unverified. A continuous stroke must be replaced with an available catalog size and reevaluated before purchase.

Removed screens: diagnostics only

- `actuator_body_margin`: User removed this constraint; the reserve is diagnostic only and does not affect optimization or feasibility. Actual actuator packaging remains unverified.
- `dig_froude_margin`: User removed the uncalibrated speed cap; constant capture/lift response assumes no additional high-speed loss.
- `dump_froude_margin`: User removed the uncalibrated speed cap; actual dumping/retention speed response remains unverified.
- `chassis_clearance_margin`: Legacy filled rectangular envelope does not represent the open plate frame; actual frame/equipment interference remains unverified.
- `dig_traction_margin`: Inactive compatibility diagnostic. User assumes exact equal/opposite cutting reactions; excavation feed traction reserve removed.
- `dig_reach_margin`: Compatibility alias of operating_reach_margin; use the active operating_reach_margin and axle_ground_margin checks.

06 / MDO ONLY

Optimization study

The candidate is separate from a released design baseline. Optimizer status: **False**. Saved status: failed_or_infeasible.

Continuous-score objective: 1,990.53 → 2,142.49 points. Complete-cycle estimate: 2,087.45 points.

Changed input	Starting value	Candidate value
Lin act side arm offset	-50.8 mm	-74.7 mm
Lin act top arm offset	60.0 mm	46.6 mm
Drum outer diameter	359.0 mm	347.9 mm
Width of each drum	340.0 mm	343.5 mm
Excavation drum RPM	30.0 rpm	18.2 rpm
Straight excavation feed speed	30.0 mm/s	30.3 mm/s
Dumping speed	60.0 rpm	59.8 rpm
Drum reduction ratio	60.0	59.7
Arm wall thickness	1.59 mm	1.00 mm
Battery capacity	259.2 Wh	220.7 Wh
Travel actuator length	470.0 mm	480.2 mm
Deposition actuator length	403.1 mm	414.8 mm
Compact actuator length	455.3 mm	469.7 mm
Maximum-extension actuator length	606.3 mm	585.5 mm
Actuator pivot height	370.0 mm	350.0 mm
Actuator pivot longitudinal position	80.0 mm	71.3 mm
Arm pivot height	230.0 mm	234.8 mm
Arm length	558.8 mm	589.4 mm
Arm pivot longitudinal position	110.0 mm	92.9 mm
Wheelbase	550.0 mm	603.6 mm
Wheel-center track	590.0 mm	597.2 mm
Aluminum arm box width	50.8 mm	47.4 mm
Aluminum arm box height	50.8 mm	48.3 mm
Aluminum cross-tube wall thickness	3.12 mm	2.77 mm
Plate strip width	18.8 mm	13.3 mm
Rocker box width	50.8 mm	49.7 mm
Rocker box height	50.8 mm	48.8 mm
Rocker wall	1.59 mm	1.01 mm
Axle outer diameter	33.0 mm	35.6 mm
Axle wall	1.00 mm	1.11 mm
Rocker pivot above wheel centers	96.2 mm	132.3 mm

APPENDIX / NATIVE UNITS

Design inputs & output data

154 inputs supplied by file; 0 CLI overrides; 0 remaining inputs use model defaults. Legacy breakout_force is inactive; actuator loads use gravity only. Optimizer-selected design values appear in the bounds table. Frozen design variables (held fixed): wheel_diameter, wheel_width. Values below retain their native model units.

All effective inputs & model assumptions

Includes mission, soil, material properties and inactive legacy inputs. Source SHA-256: `08740acfe90dadc90249b3077cb0be50e83f2750fc692e37db1f010528630dc9`.

Input port	Effective value	Native units
lin_act_side_arm_offset	-0.074696458	m
lin_act_top_arm_offset	0.046602938	m
drum_diameter	0.34791832	m
drum_length	0.34348093	m
drum_rpm	18.188136	rpm
feed_speed	0.030299009	m/s
dump_rpm	59.776794	rpm
drum_gear_ratio	59.673494	—
inlet_depth	0.025	m
arm_wall	0.001	m
battery_capacity	220.66777	W*h
travel_actuator_length	0.48019992	m
deposition_actuator_length	0.41476087	m
actuator_retracted_length	0.46969352	m
actuator_extended_length	0.58547913	m
top_pivot_height	0.35	m
actuator_pivot_x	0.071287196	m
bottom_pivot_height	0.23482544	m
arm_length	0.58943709	m
pivot_x	0.09290125	m
wheelbase	0.60355609	m
track_width	0.59725	m
arm_box_width	0.047429432	m
arm_box_height	0.048274083	m
cross_tube_span	0.16	m
cross_tube_wall	0.0027741536	m
plate_spoke_width	0.013317659	m
rocker_box_width	0.049738588	m
rocker_box_height	0.048753229	m
rocker_wall	0.0010105692	m
axle_outer_diameter	0.035631502	m
axle_wall	0.0011100047	m

Input port	Effective value	Native units
rocker_pivot_offset	0.13234864	m
minimum_ground_clearance	0.1524	m
clearance_radius_deflection_fraction	0.2	—
grouser_height	0.018	m
tipping_radius_deflection_fraction	0.2	—
tipping_reference_fill_fraction	0.9	—
wheel_compression_hard_stop	0.064	m
differential_bar_clearance_allowance	0.0254	m
empty_slope_degrees	30	deg
loaded_slope_degrees	20	deg
gravity	9.80665	m/s**2
run_time	900	s
traversal_time_one_way	45	s
travel_speed	1	m/s
acceleration	0.4	m/s**2
turn_angle	3.1415927	rad
overhead_time	8	s
electronics_power	25	W
actuator_idle_power	2	W
soil_density	1500	kg/m**3
berm_density	1500	kg/m**3
fill_fraction	0.9	—
delivery_efficiency	0.9	—
capture_fraction	0.65	—
cut_pressure	18000	Pa
internal_lift_factor	0.15	—
vertical_cut_fraction	0.5	—
force_imbalance	0.15	—
dump_revolutions	3	—
drum_efficiency	0.7	—
drum_gear_efficiency	0.8	—
drive_efficiency	0.72	—
drive_gear_efficiency	0.8	—
motor_rated_rpm	3000	rpm
traction_coefficient	0.6	—
rocker_load_factor	1.2	—
wheel_deflection_fraction	0.07	—
rolling_base	0.04	—
rolling_pressure_scale	200000	Pa
allowable_ground_pressure	25000	Pa
scrub_coefficient	0.25	—
turn_speed_fraction	0.35	—
turn_accel_fraction	0.5	—
chassis_height	0.35	m
drum_pair_gap	0.06	m
fixed_mass	6	kg

Input port	Effective value	Native units
wheel_areal_mass	4	kg/m**2
drum_areal_mass	5	kg/m**2
motor_mass_per_watt	0.006	kg/W
gear_mass_per_torque	0.05	kg/(N*m)
drive_fixed_mass	0.35	kg
drive_motor_cg_offset	0.05842	m
drum_fixed_mass	0.5	kg
battery_specific_energy	140	W*h/kg
battery_usable_fraction	0.8	—
battery_c_rate	6.667	1/h
mass_score_weight	1	—
energy_score_weight	1	—
transport_clearance	0.08	m
mechanism_clearance	0.015	m
minimum_pivot_spacing	0.08	m
minimum_stroke	0.04	m
actuator_housing_allowance	0.04	m
pit_depth	0.45	m
actuator_efficiency	0.65	—
actuator_safety_factor	1	—
breakout_force	100	N
actuator_fixed_mass	0.6	kg
actuator_mass_per_force	0.0004	kg/N
actuator_mass_per_power	0.003	kg/W
actuator_mass_per_length	1	kg/m
stored_soil_density	1500	kg/m**3
usable_volume_fraction	0.9	—
scoops_per_revolution	4	—
max_dig_froude	0.3	—
max_dump_froude	0.2	—
drum_spinup_time	2	s
drum_torque_service_factor	1.25	—
minimum_dig_depth	0.005	m
minimum_payload	0.05	kg
fresh_site_distance	0.8	m
site_speed	0.1	m/s
available_excavation_area	12	m**2
tube_density	2700	kg/m**3
tube_youngs_modulus	6.9e+10	Pa
tube_yield_strength	2.4e+08	Pa
structure_safety_factor	2	—
structure_load_factor	1.5	—
allowable_deflection	0.005	m
joint_mass	2	kg
differential_mass	1	kg
obstacle_height	0.3	m

Input port	Effective value	Native units
slope_angle	0.52359878	rad
frame_tube_diameter	0.038	m
frame_wall	0.003175	m
arm_tube_diameter	0.05	m
plate_thickness	0.00635	m
cross_tube_inner_diameter	0.03175	m
wheel_rocker_gap	0.0127	m
plate_hub_radius	0.03	m
plate_cutout_clearance	0.005	m
frame_mount_clearance	0.003	m
minimum_axle_overhang	0.01	m
plate_buckling_k	1	—
cross_tube_density	2700	kg/m**3
cross_tube_youngs_modulus	6.9e+10	Pa
cross_tube_yield_strength	2.4e+08	Pa
drive_motor_power	835.4	W
drum_motor_power	835.4	W
drive_gear_ratio	100	—
motor_bus_voltage	12	V
x44_motor_mass	0.34019428	kg
drive_gearbox_mass	0.697	kg
drum_gearbox_mass	0.753	kg
drive_stator_current_limit	0	A
drum_stator_current_limit	0	A
actuator_force_rating	4003.3995	N
actuator_speed	0.009906	m/s
actuator_catalog_stroke	0.2032	m
actuator_installation_allowance	0	m
wheel_diameter	0.355	m
wheel_width	0.152	m

Design variables and bounds

Optimizer search ranges and evaluated design variables. These ranges guide MDO and input validation; they are not all hardware requirements.

Port	Lower	Evaluated	Upper	Native units
lin_act_side_arm_offset	-0.5	-0.0746965	-0.0508	m
lin_act_top_arm_offset	-0.15	0.0466029	0.15	m
drum_diameter	0.16	0.347918	0.359	m
drum_length	0.12	0.343481	0.359	m
drum_rpm	5	18.1881	120	rpm
feed_speed	0.0001	0.030299	0.5	m/s
dump_rpm	5	59.7768	60	rpm
drum_gear_ratio	9	59.6735	125	—
inlet_depth	0.003	0.025	0.025	m
arm_wall	0.001	0.001	0.005	m
battery_capacity	10	220.668	300	W*h
travel_actuator_length	0.403098	0.4802	0.606298	m
deposition_actuator_length	0.403098	0.414761	0.606298	m
actuator_retracted_length	0.403098	0.469694	0.606298	m
actuator_extended_length	0.403098	0.585479	0.606298	m
top_pivot_height	0.35	0.35	0.7	m
actuator_pivot_x	0.02	0.0712872	0.3	m
bottom_pivot_height	0.09	0.234825	0.3	m
arm_length	0.3	0.589437	0.7	m
pivot_x	0.05	0.0929013	0.4	m
wheelbase	0.38	0.603556	0.65	m
track_width	0.5	0.59725	0.699	m
arm_box_width	0.025	0.0474294	0.1	m
arm_box_height	0.025	0.0482741	0.1	m
cross_tube_span	0.16	0.16	0.46	m
cross_tube_wall	0.001	0.00277415	0.006	m
plate_spoke_width	0.012	0.0133177	0.07	m
rocker_box_width	0.02	0.0497386	0.065	m
rocker_box_height	0.025	0.0487532	0.09	m
rocker_wall	0.001	0.00101057	0.005	m
axle_outer_diameter	0.025	0.0356315	0.09	m
axle_wall	0.001	0.00111	0.006	m
rocker_pivot_offset	0.02	0.132349	0.15	m
wheel_diameter (fixed)	0.18	0.355	0.359	m
wheel_width (fixed)	0.05	0.152	0.359	m

All output ports

Output port	Value	Native units
dry_mass	41.474134	kg
payload_mass	116.36522	kg
wheel_mass	5.8796791	kg
drum_mass	13.497812	kg
drive_mass	4.1487771	kg
battery_mass	1.5761984	kg
rover_length	1.4960501	m
rover_width	0.74925	m
rover_height	0.74925	m
ground_pressure	17881.804	Pa
actual_travel_speed	1	m/s
dig_time	167.39277	s
travel_time	52.057994	s
dump_time	10.022404	s
turn_time	4.4304517	s
cycle_time	289.95109	s
cycle_energy	11.451028	W*h
cycles	3.1039718	—
berm_volume	0.21671662	m**3
run_energy	35.543669	W*h
average_power	142.17468	W
peak_power	635.65756	W
mass_points	1532.7676	—
energy_points	609.71933	—
construction_points	2142.487	—
no_load_speed	1.4419647	m/s
loaded_motor_speed_limit	1.0234332	m/s
loaded_power_speed_limit	5.9378282	m/s
stall_drive_force	7415.6297	N
actuation_time	16.263489	s
lowering_time	10.627823	s
arm_energy	5228.6946	J
arm_peak_power	336	W
deposition_reach_margin	9.9999971e-05	m
deposition_position_margin	0.17071827	m
deposition_clearance_margin	0.53266541	m
required_actuator_force	3825.7033	N
linkage_mass	5.7099374	kg
stroke	0.10527922	m
stowed_x	0.57406587	m
stowed_y	0.57529084	m
travel_x	0.60349906	m
travel_y	0.52931816	m
deposition_x	0.30015422	m

Output port	Value	Native units
deposition_y	0.78662458	m
extended_x	0.62585128	m
extended_y	-0.016968825	m
sweep_x_min	0.30015422	m
sweep_x_max	0.68233834	m
actuator_leverage_max	5.9943039	—
actuator_attachment_distance	0.51684597	m
arm_physical_length	0.58943709	m
arm_extension_mass	0	kg
actuator_body_margin	0.011662867	m
chassis_clearance_margin	-0.12007424	m
stowed_length	1.4960501	m
stowed_height	0.74925	m
deployed_length	1.7125619	m
deployed_height	0.96058374	m
route_distance	18	m
route_turn_angle	360	deg
route_translation_time	37.268827	s
route_turn_time	14.789167	s
route_peak_acceleration	0.4	m/s**2
deposition_actuation_time	13.152399	s
deposition_arm_energy	4349.469	J
deposition_arm_peak_power	336	W
payload_volume	0.077576816	m**3
fresh_volume	0.11934895	m**3
storage_capacity	0.077576816	m**3
reachable_soil_mass	116.36522	kg
target_payload_mass	116.36522	kg
trench_depth	0.05	m
excavation_footprint	3.8621485	m**2
cut_force	184.41628	N
drum_froude	0.064351732	—
dig_overlap_time	0	s
dig_power	185.57422	W
dump_power	135.14218	W
dig_energy	16698.965	J
dump_energy	530.95821	J
drum_power_margin	1189.3617	W
drum_torque_margin	260.46444	N*m
dump_torque_margin	87.462976	N*m
dig_froude_margin	0.23564827	—
dump_froude_margin	-0.49510291	—
penetration_margin	314.51417	N
dig_traction_margin	118.98263	N
cut_depth_margin	0.01848979	m
dig_reach_margin	0.024361106	m
payload_margin	116.31522	kg
structure_mass	4.6617299	kg

Output port	Value	Native units
rocker_mass	0.56862245	kg
arm_linear_mass	0.50599898	kg/m
arm_envelope_radius	0.03383762	m
frame_section_margin	0.0077978709	m
arm_section_margin	0.010857358	m
structure_stress	1.1990001e+08	Pa
structure_stress_margin	99988.852	Pa
structure_deflection	0.0026495001	m
structure_deflection_margin	0.0023504999	m
structure_buckling_margin	2937.8224	N
empty_yaw_inertia	8.7603832	kg*m**2
empty_cg_x	5.3538094e-18	m
empty_cg_z	0.37665793	m
loaded_yaw_inertia	57.902275	kg*m**2
loaded_cg_x	0	m
loaded_cg_z	0.44645512	m
half_yaw_inertia	33.331329	kg*m**2
half_cg_x	0.35234094	m
half_cg_z	0.43193137	m
loaded_sprung_mass	147.24228	kg
loaded_sprung_cg_x	0	m
loaded_sprung_cg_z	0.46365473	m
deposition_empty_yaw_inertia	4.5524463	kg*m**2
deposition_empty_cg_x	-1.0707619e-17	m
deposition_empty_cg_z	0.47616316	m
deposition_loaded_yaw_inertia	21.796471	kg*m**2
deposition_loaded_cg_x	0	m
deposition_loaded_cg_z	0.66229743	m
deposition_half_yaw_inertia	13.174459	kg*m**2
deposition_half_cg_x	0.17523908	m
deposition_half_cg_z	0.62356569	m
relocation_time	18.631581	s
relocation_energy	256.92413	J
relocation_peak_power	20.124054	W
fresh_site_margin	0.053038135	m
site_drive_margin	156.7402	N
site_turn_time	10.381581	s
mobility_energy	6909.9147	J
drive_peak_power	610.65756	W
turn_peak_power	228.28463	W
ground_pressure_margin	7118.1963	Pa
straight_motor_margin	412.57609	N
turn_motor_margin	5322.6254	N
straight_traction_margin	169.62584	N
turn_traction_margin	0.30000041	N
straight_power_margin	2239.3556	W
turn_power_margin	2514.6641	W
mobility_stability_margin	0.28159616	m

Output port	Value	Native units
trench_traverse_margin	0.305	m
tip_stability_margin	0.12653896	m
tip_worst_mask	3	—
obstacle_pose_margin	0.28064423	m
obstacle_tip_margin	0.20896292	m
obstacle_pose_valid	1	—
obstacle_evaluated_height	0.3	m
obstacle_worst_wheel	0	—
obstacle_chassis_pitch	0.30566806	rad
obstacle_chassis_roll	0.24560148	rad
obstacle_loaded_cg_x	-0.041193719	m
obstacle_loaded_cg_y	-0.031740103	m
obstacle_loaded_cg_z	0.46103901	m
obstacle_drum_wheel_gap_bound	0.033879009	m
obstacle_drum_wheel_margin	0.018879009	m
obstacle_clearance_pose_valid	1	—
obstacle_clearance_evaluated_height	0.3	m
slope_pitch_tip_margin	0.037170472	m
slope_roll_tip_margin	0.033438545	m
slope_tip_margin	0.033438545	m
slope_tip_angle	0.57697696	rad
rock_sweep_tip_margin	0.20896292	m
rock_sweep_worst_height	0.3	m
rock_sweep_raised_wheel	0	—
rock_sweep_pitch	0.30566806	rad
rock_sweep_roll	0.24560148	rad
rock_sweep_cg_x	-0.041193719	m
rock_sweep_cg_y	-0.031740103	m
rock_sweep_cg_z	0.46103901	m
rock_sweep_wheel_load_margin	146.17099	N
rock_sweep_load_height	0.3	m
rock_sweep_load_raised_wheel	0	—
rock_sweep_unloaded_wheel	0	—
rock_sweep_pose_valid	1	—
rock_sweep_evaluated_height	0.3	m
rock_sweep_samples	41	—
rock_sweep_force_balance_error	2.7000624e-13	N
empty_slope_slip	1	—
empty_slope_solution_valid	0	—
empty_slope_traversal_margin	-0.34454036	—
empty_slope_force_shortfall	30.339522	N
empty_slope_sinkage	0.0062593767	m
empty_slope_transfer_flag_count	0	—
loaded_slope_slip	1	—
loaded_slope_solution_valid	0	—
loaded_slope_traversal_margin	-0.12539297	—
loaded_slope_force_shortfall	45.596868	N
loaded_slope_sinkage	0.01629863	m

Output port	Value	Native units
loaded_slope_transfer_flag_count	3	—
bottom_pivot_ground_clearance	0.15527628	m
rocker_pivot_ground_clearance	0.23113289	m
ground_clearance	0.15527628	m
ground_clearance_margin	0.0028762831	m
bottom_pivot_axle_radius	0.018649154	m
rocker_pivot_axle_radius	0.017815751	m
tipping_reference_mass	157.83936	kg
tipping_wheel_outer_radius	0.1775	m
tipping_wheel_tread_radius	0.1595	m
tipping_wheel_deflection	0.0355	m
tipping_wheel_deflection_ratio	0.2	—
tipping_wheel_support_height	0.142	m
tipping_loaded_cg_z	0.41095512	m
wheel_spring_compliance	9.1738654e-05	m/N
slope_wheel_load_margin	0.49999999	N
wheel_equilibrium_margin	0.8528906	—
slope_worst_heading	1.5707963	rad
slope_load_worst_heading	0.79966278	rad
slope_contact_limit_angle	0.52448788	rad
slope_equilibrium_valid	1	—
slope_pose_error	2.434164e-14	m
rock_equilibrium_valid	1	—
rock_pose_error	4.0592529e-16	m
slope_fl_normal_load	832.05353	N
slope_fl_compression	0.064	m
slope_fl_stop_reaction	134.41957	N
slope_fr_normal_load	246.1616	N
slope_fr_compression	0.022582534	m
slope_fr_stop_reaction	0	N
slope_rl_normal_load	261.78423	N
slope_rl_compression	0.024015733	m
slope_rl_stop_reaction	0	N
slope_rr_normal_load	0.49999999	N
slope_rr_compression	4.5869327e-05	m
slope_rr_stop_reaction	0	N
obstacle_equilibrium_valid	1	—
excavation_lowering_time	8.1685958	s
excavation_raising_time	8.0948927	s
operating_reach_margin	0.024361106	m
rear_dig_x	-0.67181813	m
rear_dig_actuator_length	0.56111803	m
feed_margin	1.1999997e-05	m
front_dig_y	0.14895916	m
excavation_domain_margin	0.14895916	m
front_dig_depth	0.025	m
feed_distance	3.4746892	m
rear_dig_depth	0.05	m

Output port	Value	Native units
inlet_width	0.34348093	m
front_dig_actuator_length	0.55661633	m
feed_per_scoop	0.024988	m
rear_dig_y	0.12395916	m
cut_depth	0.025	m
front_dig_x	0.67605052	m
feed_drive_distance	4.8225578	m
axle_ground_margin	0.016968825	m
feed_speed_limit	0.03031356	m/s
feed_time	159.39277	s
pair_cutting_time	114.83145	s
soil_supply_rate	0.0010407132	m**3/s
scoop_throughput	0.001041213	m**3/s
feed_drive_energy	1537.2818	J
feed_drive_peak_power	13.262381	W
feed_motor_margin	7134.7594	N
feed_power_margin	2669.4801	W
nasa_force_source_valid	0	—
nasa_tip_feed_ratio	10.935425	—
outbound_travel_time	26.028997	s
return_travel_time	26.028997	s
outbound_route_acceleration	0.4	m/s**2
return_route_acceleration	0.4	m/s**2
outbound_straight_speed_limit	1	m/s
return_straight_speed_limit	1	m/s
outbound_bwr_slip	0.22879351	—
return_bwr_slip	0.25670922	—
outbound_bwr_sinkage	0.014469278	m
return_bwr_sinkage	0.0058184317	m
outbound_bwr_transfer_flag_count	3	—
return_bwr_transfer_flag_count	0	—
pit_depth_margin	0.25907201	m
transport_clearance_margin	0.32133168	m
wheel_first_margin	0.00035900007	m
startup_position_margin	-0.010506401	m
dig_position_margin	0.11578562	m
pivot_spacing_margin	0.037185099	m
pivot_mount_margin	0.355	m
pivot_longitudinal_margin	0.2088768	m
bottom_mount_margin	0.057325437	m
stroke_margin	0.065279216	m
triangle_retracted_margin	0.06553905	m
triangle_extended_margin	0.033551931	m
toggle_margin	0.19225297	m
wheel_clearance_margin	0.000300001	m
deployed_height_margin	1.5394163	m
actuator_pivot_longitudinal_margin	0.23049085	m
actuator_toggle_margin	0.21190561	m

Output port	Value	Native units
actuator_catalog_retracted_length	0.403098	m
actuator_catalog_extended_length	0.606298	m
actuator_unit_mass	2.5567141	kg
actuator_stroke_domain_margin	0.1524	m
actuator_force_margin	177.6962	N
actuator_curve_margin	177.6962	N
actuator_curve_valid	1	—
actuator_loaded_speed	0.009996193	m/s
actuator_raising_current	13.556137	A
actuator_lowering_current	14	A
actuator_raising_time	10.531931	s
rocker_length	0.54005609	m
plate_mass_total	0.45287612	kg
cross_tube_mass_total	0.51993272	kg
suspension_axle_mass	0.12029859	kg
suspension_bearing_span	0.37011141	m
cross_tube_outer_diameter	0.037298307	m
plate_geometry_margin	0.0070361571	m
frame_packing_margin	0.091880706	m
plate_geometry_valid	1	—
structure_cg_z	0.30750866	m
structure_sprung_cg_z	0.30718358	m
structure_yaw_inertia	0.043659252	kg*m**2
plate_stress	96287204	Pa
plate_stress_margin	23712796	Pa
plate_buckling_margin	2937.8224	N
cross_tube_bending_stress	94886238	Pa
cross_tube_bending_margin	25113762	Pa
frame_mount_relative_deflection	0.00090805465	m
rocker_deflection	0.00096664402	m
axle_deflection	0.00033790974	m
arm_deflection	0.002649333	m
rocker_buckling_margin	86955.955	N
arm_buckling_margin	11268.242	N
frame_gravity_actuator_force	5738.5549	N
frame_nominal_wheel_reaction	386.96883	N
frame_force_balance_error	8.342916e-12	N
frame_load_solution_valid	1	—